System Requirements Specification

Multi-Platform Shorts Automation & AI Insights Engine.

This document consolidates all functional and non-functional requirements for the system, which publishes to three platforms: YouTube, Instagram and Facebook. It serves as a unified reference combining technical specifications, publishing pipeline logic, analytics workflows, system architecture, security, and operational constraints.

1. Functional Requirements (FR)

ID	**Module / Requirement Name**	**Detailed Description**
FR-01	Media Ingestion & File Monitoring	Automatically detect new .mp4 video uploads to the designated 'Pending_Shorts' folder in Google Drive. Trigger pipeline on detection.
FR-02	Input Validation & Integrity	Validate video container formats, file accessibility, and upload integrity before initiating automation pipelines.
FR-03	Metadata Synchronization	Fetch corresponding title, caption, category, and hashtag parameters mapped in the central Google Sheets database (Metadata DB).
FR-04	Multi-Platform Direct Publishing	Stream media streams directly and publish content across supported short-form platforms: YouTube Shorts (YouTube Data API v3 with pre-configured visibility settings); Instagram Reels (Instagram Graph API with polling verification); Facebook Reels (Meta Page Video endpoints).
FR-05	Dynamic Caption & Formatting Adaptation	Adapt text descriptions, hashtag structures, character counts, and dynamic caption formatting to conform with each target platform's specific constraints.
FR-06	State Logging & Tracking Update	Record publication outcomes (Success, Failed, or Pending) with post identifiers, direct URLs, and per-platform status columns (yt_status, ig_status, fb_status) into the central Google Sheets database.

FR-07	File Lifecycle & Archiving Cleanup	Relocate successfully processed .mp4 video files from 'Pending_Shorts' to the 'Published_Shorts' archive directory upon successful cross-posting.
FR-08	Fault Capture & Notification	Capture upload errors, record exception details, and notify system maintainers via Telegram alerts or automated email summaries whenever an API rejects an upload attempt.
FR-09	Scheduled Analytics Retrieval Trigger	Execute automated performance queries at scheduled intervals (24 hours and 72 hours post-publishing).
FR-10	Multi-Platform Metrics Extraction	Retrieve metrics (Views, Likes, Watch Time/Retention, Shares, Comments, Reach, Impressions, Saves) across YouTube and Meta (Instagram/Facebook) APIs.
FR-11	Centralized Analytics Aggregation	Consolidate raw multi-platform engagement counters into uniform tabular rows inside Google Sheets.
FR-12	Derived Performance & Virality Metrics	Calculate composite performance indicators including Hook & Retention Efficiency (comparison against opening 3-second retention drops), Like-to-View ratios, and Share-to-View viral coefficients.
FR-13	AI Context Dispatch & Prompting	Format multi-platform metrics into structured analytical prompts and route them to LLMs via OpenRouter/Gemini APIs (using prompt stored in a Set node or dedicated Sheet cell).
FR-14	AI Diagnostics & Optimization Synthesis	Generate comprehensive evaluations analyzing hook effectiveness, caption/hashtag keyword performance, posting timestamp variations (optimal publishing hours), and future CTA recommendations.
FR-15	Brief Delivery & Strategy Action Plan	Synthesize performance diagnoses and concise bulleted action plans for future videos, delivering briefs directly to the creator via Telegram alerts or automated email summaries.
FR-16	Insights & Diagnostic Storage	Save generated AI insights, performance diagnostics, and analytics history permanently into Google Sheets.

2. Non-Functional Requirements (NFR)

Architecture & Workflow Orchestration

- **NFR-01 (Automation Engine):** Built as an automated pipeline utilizing n8n as the primary workflow engine.
- **NFR-02 (Sub-workflow Modularization):** Modularized into dedicated sub-workflows: Publish-YT, Publish-IG, Publish-FB, Analytics, and AI-Report.
- **NFR-03 (Version Pinning):** Pin API versions in HTTP nodes to ensure stability and prevent breaking changes from upstream updates.
- **NFR-04 (Version Control & Documentation):** Workflows must be exported to Git repository alongside a comprehensive README.

Performance & Operational Target

- **NFR-05 (Concurrent Publishing):** Publishing pipeline must branch out and execute cross-posting to YouTube, Instagram, and Facebook concurrently.
- **NFR-06 (Polling & Status Verification):** The Instagram workflow must implement a polling loop (Wait -> Check Status -> Repeat) to confirm posts are live before finalizing.
- **NFR-07 (End-to-End SLA Target):** Complete publishing across all 3 platforms within 10 minutes of the file landing in Google Drive.
- **NFR-08 (Minimal Manual Effort):** Requires ~5 seconds of manual creator effort per video (dropping file into Drive).
- **NFR-09 (Delayed Analytics Execution):** Analytics retrieval executed in scheduled batches at 24h and 72h post-publishing.

Availability, Reliability & Fault Tolerance

- **NFR-10 (Idempotent Retries):** Log publish state per platform (yt_status, ig_status, fb_status). On retry, the workflow publishes only to platforms not yet marked 'Done', preventing duplicate posts.
- **NFR-11 (Error Workflows & Alerts):** Automatic retry on node failure in n8n, connected to an Error Workflow that dispatches instant Telegram/Email alerts.

Security & Access Control

- **NFR-12 (Credential Isolation):** Store all secrets, OAuth2 tokens, Client Secrets, Long-Lived Tokens, and Refresh Tokens securely inside n8n's Credentials Manager. Never hardcode credentials in nodes or exported workflow JSON files.
- **NFR-13 (Least Privilege Access):** Apply least-privilege principles—upload-only permissions for content delivery endpoints and read-only permissions for insights/analytics APIs.

- **NFR-14 (Service Account Scope):** Use dedicated Google Service Account restricted strictly to the designated Drive folders and Sheet.

Cost Optimization

- **NFR-15 (Free-Tier Operation):** Core architecture designed to operate within the free-tier API quotas of Google Cloud, Meta Developers, and OpenRouter.

API Integration & Compatibility

- **NFR-16 (Supported Integrations):** Strict integration compatibility with YouTube Data API v3 & Analytics, Instagram Graph API, Facebook Page Video API, OpenRouter API, and Google Workspace APIs.

Scalability (Horizontal & Vertical)

- **NFR-17 (Horizontal Platform Scaling):** New social platforms (e.g., LinkedIn, X, Pinterest) can be added as independent sub-workflows without altering existing workflow code.
- **NFR-18 (Vertical Load Scaling):** Designed to scale by requesting higher API quotas (YouTube, Meta app review), migrating metadata storage from Google Sheets to PostgreSQL (to eliminate rate limits), and transitioning to paid AI models for higher rate limits.

Compliance, Privacy & Terms of Service

- **NFR-19 (Content Disclosure):** Support mandatory disclosure labels for AI-generated or branded content and avoid copyrighted audio assets.
- **NFR-20 (Data Privacy & Non-Scraping):** Extract metrics exclusively from creator's own authenticated accounts. Strictly avoid storing viewer personal data or scraping external content.
- **NFR-21 (ToS Adherence):** Strict adherence to platform terms of service and API rate limits.